Chapter 9 — Electrostatics and Current Electricity

Complete Notes · PECTAA Physics 11 · Board-style longs + conceptual shorts

Topic 1 — Coulomb’s law and electric field strength

Q.1 State Coulomb’s law and explain electric field strength with examples.

Coulomb’s law states that the force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them: $F = kq_1q_2/r^2$, where $k = 1/4\pi\varepsilon_0 \approx 9 \times 10^9$ N m^2 C^{-2} in free space. Like charges repel and unlike charges attract. In a medium of relative permittivity ε_r, the force is reduced by the factor ε_r.

Electric field strength is the force per unit positive charge: $E = F/q$. For a point charge, $E = kq/r^2$, and for a uniform field between parallel plates, $E = V/d$. The field is a vector and its direction is the direction of force on a positive charge.

Textbook check (p. 186): the first measurement of the force is dated 1874, yet the box gives Coulomb’s life as 1736–1806. His measurements were published in 1785.

Important Conceptual Short Questions — Topic 1

1. Why does the electric force decrease rapidly with distance?

Coulomb’s law is an inverse-square law, $F \propto 1/r^2$. Doubling the distance spreads the same influence over four times the area, so the force becomes one quarter.

2. How is the electric field between parallel plates uniform?

Between large oppositely charged plates the field lines are parallel and equally spaced, so the force per unit charge has the same magnitude and direction everywhere, giving $E = V/d$.

3. Why is the force reduced when charges are placed in a dielectric?

The dielectric becomes polarized and produces an opposing field. The effective force is therefore reduced by the relative permittivity ε_r of the medium.

Topic 2 — Electric flux and Gauss’s law

Q.2 Explain electric flux and Gauss’s law and apply them to a charged sphere.

Electric flux through a surface is the product of the field and the area perpendicular to the field: $\Phi = EA \cos\theta$. Gauss’s law states that the total flux through a closed surface equals the charge enclosed divided by ε_0: $\Phi = Q/\varepsilon_0$. It is useful when the charge distribution has symmetry.

For a charged conducting sphere, the field outside is the same as if all the charge were concentrated at the centre, $E = kQ/r^2$. Inside the conductor the field is zero because the charges reside on the outer surface. This result explains shielding and

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the behaviour of charged hollow conductors.

Important Conceptual Short Questions — Topic 2

1. Why is the electric field inside a charged conductor zero?

Free charges move until they reach the outer surface and arrange themselves so that their fields cancel inside. If a field remained, charges would continue to move, so equilibrium requires zero internal field.

2. How does Gauss's law simplify the field of a sphere?

A spherical Gaussian surface has the same field magnitude at every point and the field is normal to the surface. The flux is then $E \times 4\pi r^2$, which gives $E = kQ/r^2$ outside the sphere.

3. Why does the flux through a closed surface depend only on the enclosed charge?

Field lines from outside charges enter and leave the surface, giving zero net flux. Only lines beginning or ending on enclosed charge contribute to the net flux.

4. How does a Faraday cage shield its interior from external electric fields?

The external field induces charges on the conducting surface of the cage. These induced charges produce an internal field that cancels the external field inside, so the net field in the cavity is zero and sensitive equipment is protected.

Topic 3 — Electric potential, electron volt and motion of charged particles

Q.3 Explain electric potential, its relation with electric field, and the motion of a charged particle in a uniform field.

Electric potential at a point is the work done per unit positive charge in bringing it from infinity to that point: $V = W/q$. Potential difference is the work done per unit charge between two points. The electric field is the negative potential gradient: $E = -\Delta V/\Delta d$. The electron volt is the energy gained by an electron moving through a potential difference of one volt: $1\ eV = 1.6 \times 10^{-19}\ J$.

A charged particle in a uniform electric field experiences a constant force $F = qE$ and therefore a constant acceleration. A positive charge accelerates along the field and a negative charge opposite to it. If the particle enters perpendicular to the field, its path is parabolic, just like a projectile under gravity.

Textbook check (p. 196): the graph titled "ECG (Normal alpha rhythm)" is a brain recording; the caption correctly describes electroencephalography, so the heading should be EEG.

Important Conceptual Short Questions — Topic 3

1. Why is the electric field called the negative potential gradient?

The field points in the direction in which potential decreases most rapidly. The negative sign shows that a positive charge moves from higher to lower potential.

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2. How is the electron volt related to the joule?

One electron volt is the energy gained by an electron of charge 1.6×10^{-19} C moving through 1 V, so 1 eV = 1.6×10^{-19} J.

3. Why does a charged particle entering a uniform field perpendicularly follow a parabolic path?

The field gives a constant acceleration in one direction while the velocity in the perpendicular direction remains constant. This combination of uniform motion and uniformly accelerated motion produces a parabola.

Topic 4 — Electric current, Ohm's law and resistivity

Q.4 Explain current in terms of drift velocity and describe Ohm's law and resistivity.

Electric current is the rate of flow of charge, $I = Q/t$. In a conductor of cross-sectional area A with n charge carriers per unit volume, each of charge q and drift velocity v, the current is $I = nAvq$. Although the drift velocity is small, the electric field is established almost instantly, so all free electrons begin to drift together and a bulb lights at once.

Ohm's law states that the current through a conductor is proportional to the potential difference across it, provided physical conditions remain constant: $V = IR$. The resistance of a wire is $R = \rho L/A$, where ρ is the resistivity. In metals, resistivity increases with temperature because lattice vibrations scatter electrons more strongly. Electrical power is $P = VI = I^2R = V^2/R$.

Important Conceptual Short Questions — Topic 4

1. Why does a bulb light immediately although electron drift is slow?

The electric field is set up through the circuit almost at the speed of light, so free electrons everywhere start drifting at the same time. Energy is transferred by the field, not by one electron travelling from the switch to the bulb.

2. How does temperature affect the resistance of a metal?

Higher temperature increases the vibration of the lattice ions, so electrons are scattered more often. The resistivity and therefore the resistance of the metal increase.

3. Why is resistivity a property of the material while resistance is not?

Resistance depends on length and area as well as material, $R = \rho L/A$. Resistivity removes these geometric factors, so it characterises the material itself.

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Topic 5 — EMF, internal resistance and Kirchhoff's rules

Q.5 Explain emf and terminal potential difference and state Kirchhoff's rules.

The emf E of a source is the energy supplied per unit charge when no current flows. When current I flows through a source of internal resistance r, some voltage is lost inside the source, so the terminal potential difference is V = E − Ir. Thus emf and terminal voltage are equal only when no current is drawn.

Kirchhoff's first rule states that the sum of currents entering a junction equals the sum leaving it; it expresses conservation of charge. The second rule states that the algebraic sum of potential changes around a closed loop is zero; it expresses conservation of energy. When applying the loop rule, crossing a cell from negative to positive is a rise +E, and from positive to negative is −E. Crossing a resistor in the direction of current is a drop −IR, and against the current is +IR.

Textbook check (p. 210): the written sign rules for Kirchhoff's second rule are reversed. The correct rules are those used in the book's own worked example: − to + gives +E, + to − gives −E, with current gives −IR, against current gives +IR.

Important Conceptual Short Questions — Topic 5

1. Why is terminal voltage less than emf when current flows?

Current through the internal resistance causes a voltage drop Ir inside the source. The terminal voltage is therefore V = E − Ir, which is less than the emf.

2. How does Kirchhoff's first rule express conservation of charge?

Charge cannot accumulate at a junction. The total current entering must equal the total current leaving, so charge is conserved.

3. How does Kirchhoff's second rule express conservation of energy?

A charge returning to its starting point in a loop has the same potential energy. The total rises and drops around the loop must therefore cancel to zero.

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Topic 6 — Wheatstone bridge, potentiometer and sensors

Q.6 Explain the Wheatstone bridge, the potentiometer and the use of thermistors and LDRs in potential dividers.

A Wheatstone bridge balances when no current flows through the galvanometer. The balance condition is $R_1/R_2 = R_3/R_4$, so an unknown resistance can be found from three known resistances. A potentiometer compares emfs using a null method; at balance $E_1/E_2 = \ell_1/\ell_2$. It works as a potential divider, not as a Wheatstone bridge.

A thermistor is a temperature-dependent resistor; most thermistors lose resistance when heated. A light dependent resistor loses resistance when light falls on it. In a potential divider, a change in the sensor resistance changes the output potential difference, so the circuit can be used as a temperature or light sensor.

Textbook check (p. 213): the potentiometer is said to work on the principle of the Wheatstone bridge. It actually works as a potential divider using a null reading.

Important Conceptual Short Questions — Topic 6

1. Why does the galvanometer resistance not affect a balanced Wheatstone bridge?

At balance no current flows through the galvanometer, so there is no voltage drop across it. Its resistance therefore plays no part in the balance condition.

2. How does a potentiometer compare two emfs accurately?

It uses a null method: at balance no current is drawn from the cell being measured, so its internal resistance causes no voltage drop and the emf is proportional to the balancing length.

3. Why does the output of a potential divider change when an LDR is illuminated?

Light lowers the resistance of the LDR. In a potential divider this changes the fraction of the supply voltage across the LDR, so the output potential difference changes with light intensity.

4. How can carbon fibres help inspectors check a concrete bridge?

Carbon fibres embedded in the concrete conduct electricity. Cracks or damage change the resistance of the fibre network, so measuring the resistance allows inspectors to detect internal damage and check the reliability of the bridge.

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